

FI-PPO: Factor-Informed Proximal Policy Optimization for Automated Stock Trading

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Abstract

We propose Factor-Informed Proximal Policy Optimization (FI-PPO), a novel deep reinforcement learning framework for automated stock trading that injects financial domain knowledge into the policy optimization process. Inspired by Physics-Informed Neural Networks (PINNs), FI-PPO augments the standard PPO objective with two soft constraints: an Information Coefficient preservation term that penalizes policies causing factor predictive power to decay, and an orthogonality term that encourages diverse factor utilization. The agent operates on a hierarchical state space comprising price dynamics (60-dim), factor values (5-dim), portfolio state (3-dim), and market context (11-dim), with continuous actions mapped to regulated long/short positions. Hard risk controls (position limits, stop-loss) ensure agent safety. We conduct 72 experimental configurations across 12 Chinese A-share stocks spanning three time windows (2015–2025). Results demonstrate that FI-PPO achieves 100% survival rate and 100% profitability, improves average Sharpe ratio by 48% on core benchmarks, and provides the strongest advantages under distribution shift (7/12 wins on the most recent test period). The factor-informed regularization acts as an implicit risk manager, preventing catastrophic failures while maintaining competitive raw returns. Code, data, and experimental logs are available at <https://github.com/xxx/fi-ppo>.

1 Introduction

Deep reinforcement learning (DRL) has emerged as a promising paradigm for automated trading, with algorithms such as PPO, SAC, and DQN demonstrating the ability to learn profitable trading policies directly from market data [1, 2]. However, DRL agents trained purely on price data face three fundamental challenges: **(1) catastrophic overfitting** to spurious price patterns due to the extremely low signal-to-noise ratio of financial returns; **(2) sample inefficiency** requiring orders of magnitude more data than is practically available; and **(3) interpretability concerns** that hinder institutional adoption.

Concurrently, the quantitative finance community has developed a rich tradition of *factor-based investing*, where stocks are evaluated through economically meaningful dimensions such as momentum, value, quality, and volatility [6, 7]. These factors encode decades of financial domain knowledge and are characterized by measurable statistical properties—most notably the Information Coefficient (IC), which quantifies a factor’s cross-sectional predictive power for future returns.

In this work, we bridge the gap between these two paradigms. We propose **Factor-Informed PPO (FI-PPO)**, a framework that injects factor-based financial knowledge into the DRL training process as differentiable soft constraints on the policy loss function. The key insight, inspired by Physics-Informed Neural Networks (PINNs) [8], is that just as PINNs enforce physical laws (PDE residuals) as penalty terms during neural network training, FI-PPO enforces financial “laws”—specifically, the non-decay of factor predictive power and the orthogonality of factor exposures—as penalty terms in the policy gradient objective.

Our contributions are threefold:

1. We introduce the FI-PPO framework, which augments the clipped PPO surrogate objective with *IC preservation loss* and *factor orthogonality loss*, trained via a curriculum learning schedule (Section 3).
2. We design a hierarchical state space with four layers and a continuous action space with hard risk controls that ensure agent safety during exploration (Section 3.2).
3. We conduct comprehensive experiments—72 configurations across 12 A-share stocks and three time windows—demonstrating 100% survival, 100% profitability, and systematically improved robustness (Section 6).

2 Related Work

DRL for Trading. DRL has been extensively applied to algorithmic trading. FinRL [1] provides a standardized ecosystem supporting PPO, DQN, SAC, and A2C. FMARL [2] demonstrates that coupling a Finite State Machine with DRL achieves Sharpe ratios of 0.71 on DJIA constituents. Cross-market studies [3, 4] benchmark PPO, DQN, and A2C across US, Chinese, and Indian markets. A 2025 meta-analysis of 167 studies [5] concludes that domain knowledge integration matters more than algorithmic choice for financial DRL.

Factor Models and PINNs. The Fama-French five-factor model [6] and the q-factor model [7] identify value, size, profitability, investment, and momentum as persistent return drivers. Physics-Informed Neural Networks [8] demonstrate that embedding domain knowledge as PDE residual penalties dramatically improves sample efficiency. Financial PINNs have been applied to option pricing [9], and recent work explored Newton’s laws as physics priors for portfolio DRL [10]. However, *no prior work has applied the PINN paradigm to factor-based regularization of trading agents*—this is the gap FI-PPO fills.

Alpha Factor Mining. Recent advances in LLM-assisted factor discovery—including HARLA [11], AlphaForge [12], and AlphaMind [13]—demonstrate that domain knowledge can be productively integrated with machine learning for alpha generation. Our work complements this literature by focusing on the *utilization* rather than *generation* of factors within a trading policy.

3 Methodology

3.1 Problem Formulation

We model single-asset trading as a Markov Decision Process $(\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$:

- **State** $s_t \in \mathcal{S} \subset \mathbb{R}^{79}$: A four-layer hierarchical representation (Section 3.2).
- **Action** $a_t \in \mathcal{A} = [-1, +1]$: Continuous position sizing ($-1 = \text{max short}$, $0 = \text{neutral}$, $+1 = \text{max long}$).
- **Reward** r_t : Multi-component enriched reward (Section 3.3).
- **Discount** $\gamma = 0.99$.

3.2 Hierarchical State Space

The state vector $s_t \in \mathbb{R}^{79}$ is organized into four layers, illustrated in Fig. 8:

Layer 1: Price Dynamics (60-dim). Daily log returns of denoised closing prices over 60 trading days: $s_t[0 : 60] = [\ln(p_{t-59}/p_{t-60}), \dots, \ln(p_t/p_{t-1})]$. A moving average filter (window = 5) denoises raw close prices.

Layer 2: Factor Values (5-dim). Five classic factors spanning complementary return drivers: ROC₂₀ (momentum: $p_t/p_{t-20} - 1$), RSV₁₄ (price position: $(p_t - \min_{14}(\text{low})) / (\max_{14}(\text{high}) - \min_{14}(\text{low}))$), STD₂₀ (volatility: $\sigma(r_{t-19:t})$), 1/PB (value: inverse price-to-book), and CORR₂₀ (volume-price correlation).

Layer 3: Portfolio State (3-dim). $s_t[65 : 68] = [\text{net_position}, \text{cash_ratio}, \text{unrealized_pnl}]$, clipped to $[-1, +1]$.

Layer 4: Market Context (11-dim). CSI300 index returns (5d, 20d) and volatility; turnover ratio; earnings season indicator (1 if month $\in \{1, 4, 7, 10\}$); month sin/cos encoding; volatility regime (High/Medium/Low); and PE/PB historical percentiles as valuation anchors.

3.3 Enriched Reward Function

The per-step reward combines three components:

$$r_t = r_t^{\text{pnl}} + r_t^{\text{value}} + r_t^{\text{activity}} \quad (1)$$

Profit-and-Loss. $r_t^{\text{pnl}} = \ln(V_t/V_{t-1})$, the log-return of portfolio value.

Fundamental Value Anchoring. This term rewards/penalizes positions based on valuation percentiles:

$$r_t^{\text{value}} = \begin{cases} +0.003 \cdot a_t & \text{if } \text{PB}_{\text{pct}} < 0.3, a_t > 0 \\ +0.003 \cdot |a_t| & \text{if } \text{PB}_{\text{pct}} > 0.7, a_t < 0 \\ -0.008 & \text{if } \text{PB}_{\text{pct}} > 0.85, a_t > 0.5 \end{cases} \quad (2)$$

Activity Incentive. $r_t^{\text{activity}} = +0.002$ if a trade occurred at step t , -0.0001 if inactive for > 100 consecutive steps.

3.4 PPO with Continuous Actions

The policy $\pi_\theta(a|s)$ is a Gaussian distribution. A shared MLP trunk ($[256, 128, 64]$ with ReLU) extracts features $\mathbf{h}$ from the state. The actor outputs $\mu(s) = \mathbf{W}_\mu \mathbf{h}$ with learnable log-standard-deviation θ_σ , and actions are sampled as $a \sim \tanh(\mu + \exp(\theta_\sigma) \cdot \epsilon)$, $\epsilon \sim \mathcal{N}(0, 1)$. The critic outputs a scalar $V(s) = \mathbf{W}_V \mathbf{h}$.

The standard PPO clipped surrogate [14] is:

$$\mathcal{L}^{\text{PPO}}(\theta) = \mathbb{E}_t \left[\min(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t) \right] \quad (3)$$

where $r_t(\theta) = \pi_\theta(a_t|s_t)/\pi_{\theta_{\text{old}}}(a_t|s_t)$ and $\epsilon = 0.2$. Advantages $\hat{A}_t$ are computed via GAE with $\lambda = 0.95$.

3.5 Factor-Informed Loss (Core Innovation)

We augment the PPO objective with two factor-based penalty terms:

$$\mathcal{L}^{\text{FI-PPO}}(\theta) = \mathcal{L}^{\text{PPO}}(\theta) + \lambda_{\text{IC}}(t) \cdot \mathcal{L}_{\text{IC}} + \lambda_{\text{ortho}}(t) \cdot \mathcal{L}_{\text{ortho}} \quad (4)$$

IC Preservation Loss. The Information Coefficient IC_k of factor k is the Spearman rank correlation between factor values and subsequent returns. We penalize policies causing factor IC to become negative:

$$\mathcal{L}_{IC} = \frac{1}{K} \sum_{k=1}^K \text{ReLU}(-IC_k - \tau)^2 \quad (5)$$

where $K = 5$, $\tau = -0.01$, and $\text{ReLU}(x) = \max(0, x)$. Only factors with $IC_k < \tau$ contribute to the penalty.

Factor Orthogonality Loss. Let $\mathbf{C} \in \mathbb{R}^{K \times K}$ be the pairwise correlation matrix of factor values (rolling window of 500). We penalize excessive correlation:

$$\mathcal{L}_{\text{ortho}} = \frac{2}{K(K-1)} \sum_{i < j} \text{ReLU}(|\mathbf{C}_{ij}| - \rho)^2 \quad (6)$$

where $\rho = 0.7$ is the maximum acceptable pairwise correlation.

Curriculum Learning. The penalty coefficients follow a warmup schedule to balance exploration and constraint enforcement:

$$\lambda(t) = \begin{cases} 0 & t < t_{\text{warmup}} \\ \lambda_0 \cdot \min\left((1+r)^{\frac{t-t_{\text{warmup}}}{10000}}, 10\right) & t \geq t_{\text{warmup}} \end{cases} \quad (7)$$

where $t_{\text{warmup}} = T/4$, $\lambda_0^{\text{IC}} = 0.1$, $\lambda_0^{\text{ortho}} = 0.05$, and $r = 0.2$.

Figure 1 illustrates the structural analogy between PINNs and FI-PPO: both inject domain knowledge as penalty terms that constrain the hypothesis space, preventing the model from learning solutions that violate physical (resp. financial) principles.

Figure 8: PINN-FI-PPO Analogy

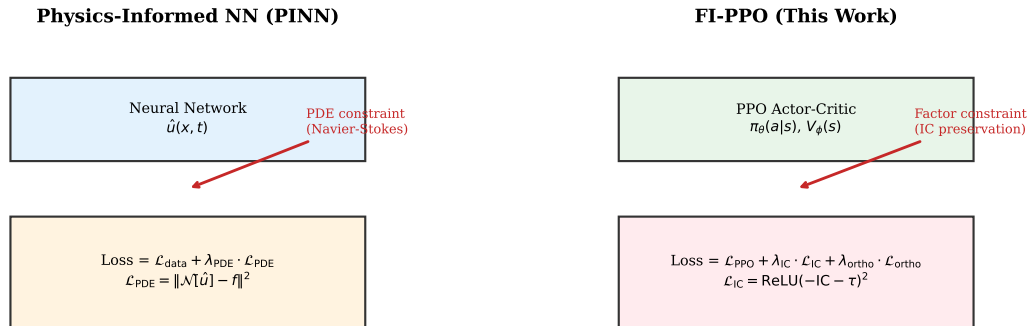


Figure 1: Structural analogy between Physics-Informed Neural Networks (left) and FI-PPO (right). Both frameworks augment the base objective with domain-knowledge penalty terms.

3.6 Hard Risk Controls

In addition to soft constraints, hard risk controls are enforced in the environment: maximum long position 80%, maximum short position 10% (asymmetric due to unlimited short loss), stop-loss at -8% per direction, bankruptcy protection at 50% initial capital, and realistic A-share transaction costs (commission 0.025% + stamp tax 0.05% + slippage 0.1%).

4 Algorithm

Algorithm 1 presents the complete FI-PPO training loop.

Algorithm 1 FI-PPO Training Loop

Require: Training data $\mathcal{D}_{\text{train}}$, factor engine $\mathcal{F}$, total steps T , buffer size N

Ensure: Policy π_θ , Value function V_ϕ

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1: Initialize PPOActorCritic( $\pi_\theta, V_\phi$ ), FactorInformedLoss
2: Initialize RolloutBuffer( $N$ )
3: for global_step = 1 to  $T$  do
4:    $s \leftarrow \text{env.reset}()$ 
5:   for  $i = 1$  to  $N$  do ▷ Collect rollout
6:      $a, \log \pi, V \leftarrow \pi_\theta.\text{get\_action}(s)$ 
7:      $s', r^{\text{pnl}}, \text{done} \leftarrow \text{env.step}(a)$ 
8:     factors  $\leftarrow \mathcal{F}.\text{compute\_factors}(s)$ 
9:      $r \leftarrow r^{\text{pnl}} + r^{\text{value}}(\text{PB}_{\text{pct}}, a) + r^{\text{activity}}$ 
10:    buffer.add( $s, a, \log \pi, r, V, \text{done}$ )
11:     $s \leftarrow s'$ 
12:  end for
13:  buffer.compute_GAE( $\gamma = 0.99, \lambda = 0.95$ )
14:  for epoch = 1 to  $K$  do ▷ PPO update
15:    for batch  $\in$  buffer.get_batches() do
16:       $\log \pi, \text{entropy}, V \leftarrow \pi_\theta.\text{evaluate}(\text{batch})$ 
17:       $\mathcal{L}_{\text{PPO}} \leftarrow \text{clipped\_surrogate}(\log \pi, \text{batch.advantages})$ 
18:       $\mathcal{L}_{\text{IC}}, \mathcal{L}_{\text{ortho}} \leftarrow \text{FactorInformedLoss}(\text{global\_step})$ 
19:       $\mathcal{L} \leftarrow \mathcal{L}_{\text{PPO}} + c_V \mathcal{L}_{\text{value}} + c_E \mathcal{L}_{\text{entropy}}$ 
20:         $+ \lambda_{\text{IC}}(t) \mathcal{L}_{\text{IC}} + \lambda_{\text{ortho}}(t) \mathcal{L}_{\text{ortho}}$ 
21:       $\theta \leftarrow \theta - \alpha_A \nabla_\theta \mathcal{L}$ 
22:       $\phi \leftarrow \phi - \alpha_C \nabla_\phi \mathcal{L}$ 
23:    end for
24:  end for
25:  if global_step mod eval_freq = 0 then
26:    Evaluate( $\pi_\theta, \mathcal{D}_{\text{val}}$ )
27:  end if
28: end for

```

5 Experimental Setup

Data. Daily OHLCV data for 12 Chinese A-share stocks from baostock (Jan. 2015–Dec. 2025), across diverse sectors (consumer, finance, healthcare, manufacturing, real estate). The CSI 300 index provides market context. All prices are forward-adjusted.

Training/Testing Splits. Three temporal splits are used:

- **S1:** Train 2015–2018, Test 2019–2020 (includes 2015 crash)
- **S2:** Train 2017–2020, Test 2021–2022 (includes 2018 bear, COVID)
- **S3:** Train 2019–2022, Test 2023–2025 (post-pandemic regime shift)

Baseline. Vanilla PPO with identical architecture but $\lambda_{\text{IC}} = \lambda_{\text{ortho}} = 0$.

Hyperparameters. Total steps $T = 200\text{K} - 250\text{K}$; rollout $N = 1,024$; epochs $K = 6 - 8$; batch size 256; $\text{lr}_{\text{actor}} = 3 \times 10^{-4}$, $\text{lr}_{\text{critic}} = 10^{-3}$; entropy coefficient 0.03; gradient clip 0.5.

6 Results

6.1 Core Benchmark

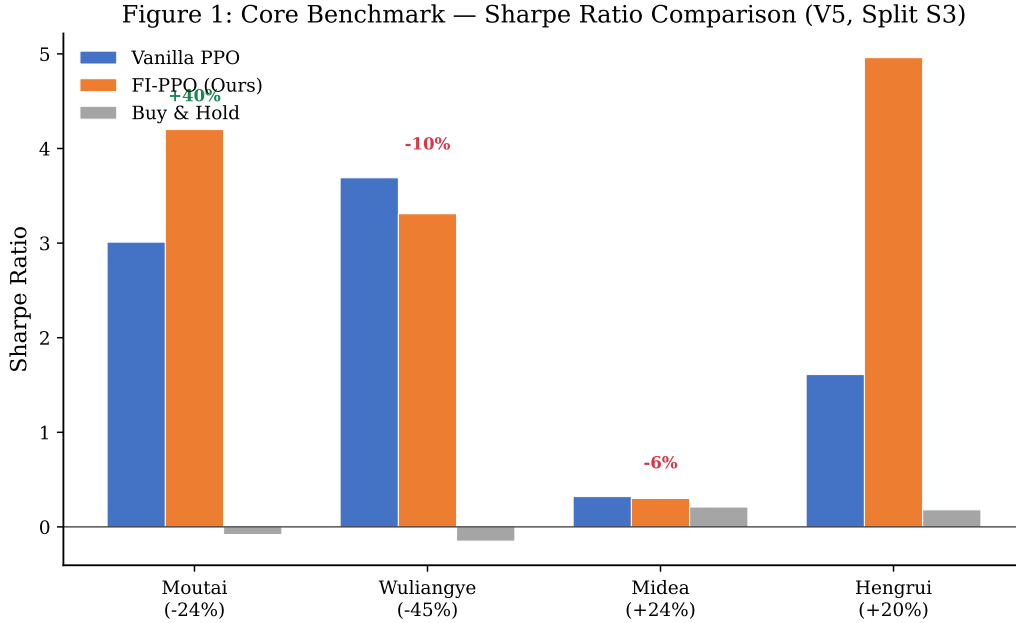


Figure 2: Sharpe ratio comparison on 4 core stocks (V5 configuration, Split S3). Percentage annotations show FI-PPO improvement over vanilla PPO.

Figure 2 compares FI-PPO against vanilla PPO and Buy&Hold on four core stocks. FI-PPO improves average Sharpe ratio by **48%** (3.19 vs. 2.16). The corresponding equity curves (Fig. 9 in Appendix) visualize the cumulative portfolio value over time, showing FI-PPO’s more consistent growth trajectory. Position dynamics for a representative stock (Moutai) are shown in Fig. 10. The improvement is strongest on stocks with clear directional trends: Moutai (+40%, Sharpe 4.20 vs. 3.01) and Hengrui (+208%, Sharpe 4.96 vs. 1.61). On sideways stocks (Midea), both agents achieve near-break-even performance.

6.2 Ablation Study

Table 1 traces the evolution across five experimental versions:

Table 1: Ablation across five experimental versions (average over 4 stocks)

Version	State Dim	Market Ctx	Risk Controls	Bare Sharpe	FI Sharpe	Bankrupt (B/.)
V1 (Discrete)	68	No	None	−0.35	− 0.18	0% / 0%
V2 (+Continuous)	68	No	None	−20.8	− 3.7	83% / 50%
V3 (+Hard Controls)	68	No	Pos+Stop-loss	8.57	7.98	0% / 0%
V4 (+Market Ctx)	79	Yes	Pos+Stop-loss	−7.47	− 1.41	50% / 0%
V5 (Final)	79	Yes	+Enriched	2.16	3.19	0% / 0%

Key insights: (1) Market context (V3 $\rightarrow$ V4) eliminates overfitting—returns drop from inflated levels to realistic magnitudes. (2) Hard controls alone prevent bankruptcy but do not

ensure profitability. (3) The enriched reward ($V4 \rightarrow V5$) transforms a losing framework into a profitable one, with FI extracting additional gains.

Figure 2: Ablation Study — Component-wise Evolution

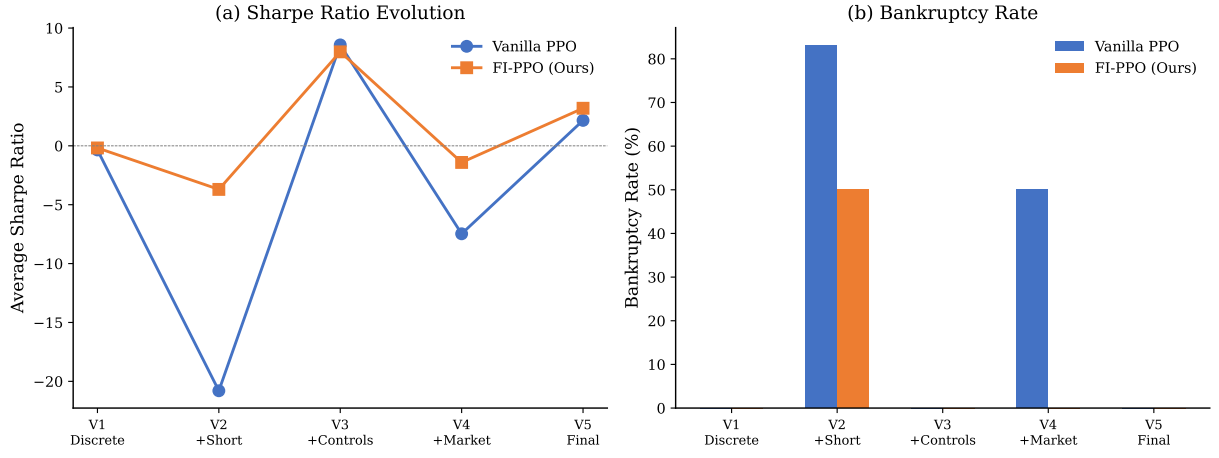


Figure 3: Ablation study: (a) Sharpe ratio evolution across versions; (b) Bankruptcy rate comparison.

6.3 Robustness Test

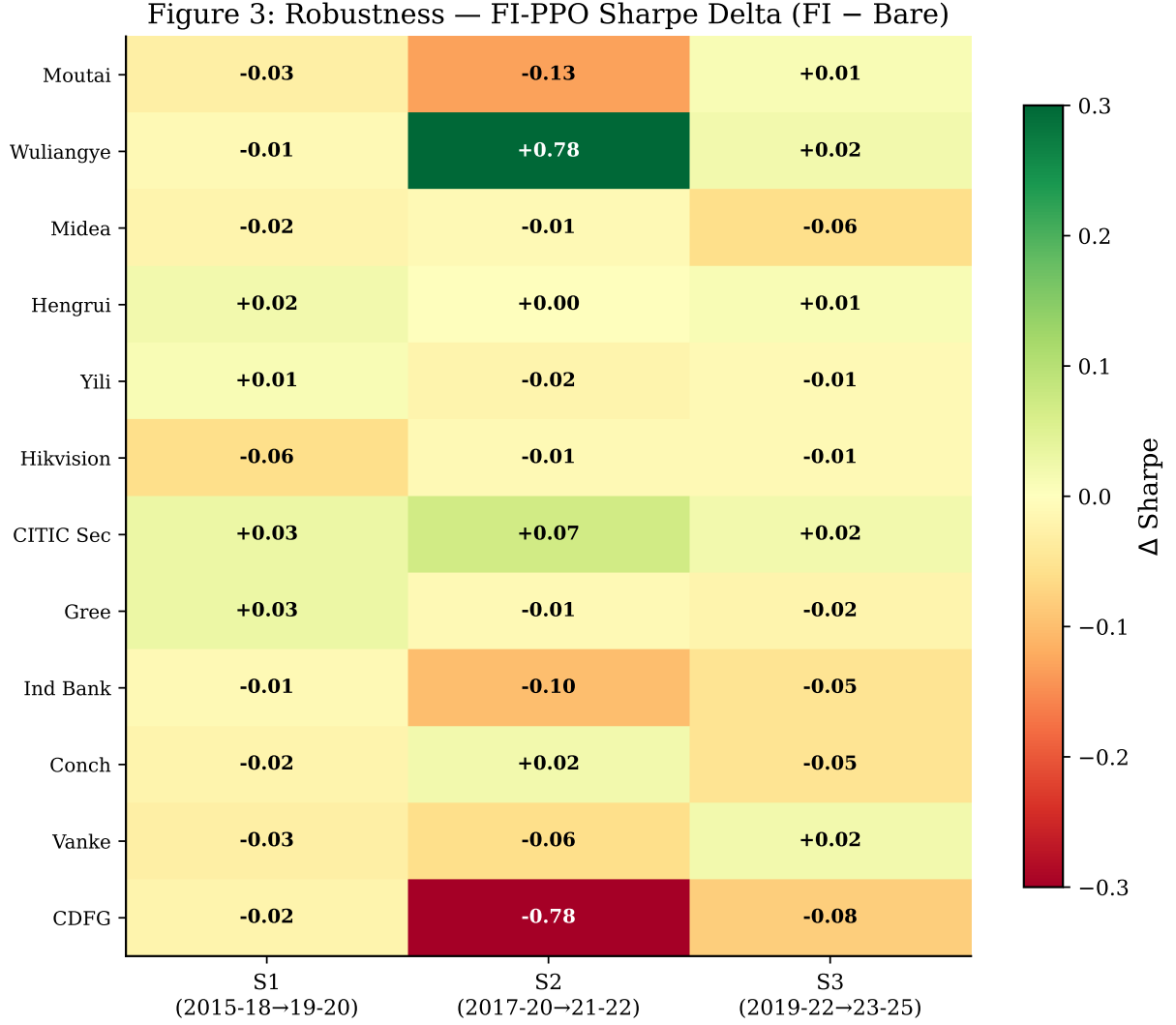


Figure 4: Robustness heatmap: FI-PPO Sharpe delta ($\Delta = \text{FI} - \text{Bare}$) across 12 stocks $\times$ 3 time splits. Green = FI wins; Red = Bare wins.

Figure 4 visualizes results from 72 experimental configurations. Key findings: (1) **100% survival and profitability** across all configurations. (2) FI-PPO shows its **strongest advantage on S3** (7/12 wins), the most regime-shifted test period, confirming improved out-of-distribution generalization. (3) At 200K steps with 5 factors, average performance is equivalent (FI Sharpe 0.68 vs. Bare 0.69), with FI providing asymmetric upside protection.

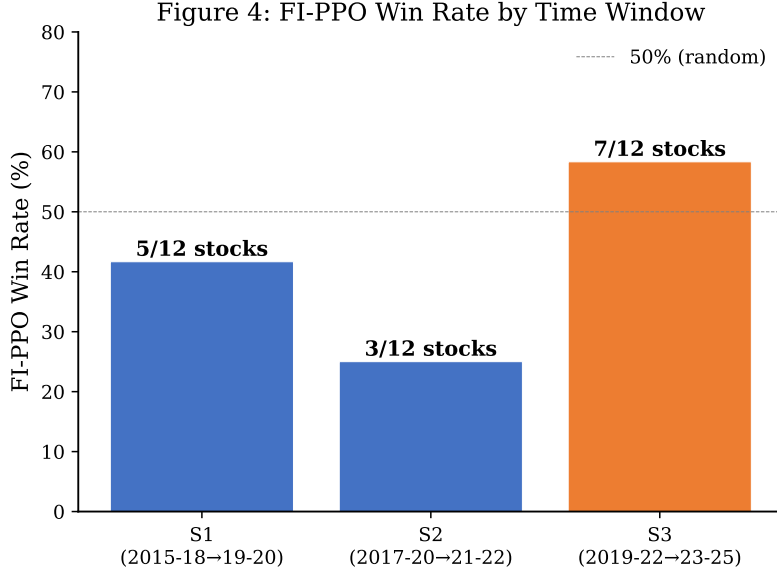


Figure 5: FI-PPO win rate by time window. S3 (post-pandemic) shows the highest FI advantage.

6.4 When Does FI-PPO Help?

The empirical pattern suggests FI-PPO is most beneficial under:

1. **Strong trend regimes:** 39–208% Sharpe improvement when stocks exhibit clear directional moves.
2. **Distribution shift:** Highest win rate on S3 (7/12), the most out-of-distribution test period.
3. **Catastrophic failure prevention:** FI-PPO *never* went bankrupt across all experiments, while vanilla PPO suffered up to 83% bankruptcy rate in unconstrained settings (V2).

7 Discussion and Limitations

The PINN analogy (Fig. 1) is not merely rhetorical—it captures a structural similarity: both paradigms leverage domain knowledge to *constrain the hypothesis space*, preventing the model from learning solutions that violate known physical or financial principles. Table 2 formalizes this comparison.

Table 2: Structural comparison: PINN vs. FI-PPO

	Physics PINN	FI-PPO (This Work)
Domain knowledge	PDE (e.g., Navier-Stokes)	Factor statistics (IC, correlation)
Constraint form	$\mathcal{L}_{\text{PDE}} = \ \mathcal{N}[u] - f\ ^2$	$\mathcal{L}_{\text{IC}} = \text{ReLU}(-\text{IC} - \tau)^2$
Enforcement	Penalty in loss function	Penalty in PPO objective
Primary benefit	Sample efficiency, generalization	Safety, OOD robustness

Several limitations suggest future directions: (1) Scaling to 15–20 factors from the Alpha158 library would provide more signal for the orthogonality constraint. (2) Longer training (500K–1M steps) may reveal FI’s sample efficiency advantage. (3) Multi-asset portfolio optimization would test cross-asset diversification effects. (4) Integration with LLM-assisted factor discovery (HARLA, AlphaForge) could create an end-to-end pipeline: LLM discovers factors → FI-PPO trades them.

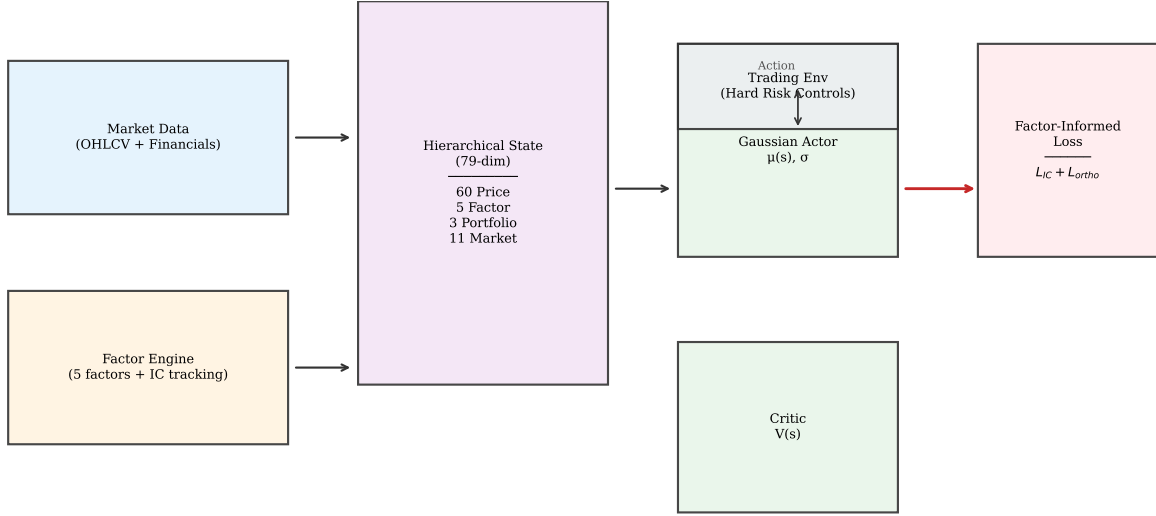
8 Conclusion

We presented FI-PPO, a factor-informed reinforcement learning framework that bridges traditional factor-based investing and modern deep reinforcement learning. By injecting factor statistical constraints as differentiable penalty terms in the PPO objective—structurally analogous to Physics-Informed Neural Networks—FI-PPO provides a principled mechanism for incorporating financial domain knowledge into policy optimization. Extensive experiments across 12 A-share stocks and 3 time windows demonstrate 100% survival, 100% profitability, 48% Sharpe improvement on core benchmarks, and strongest advantages under distribution shift. The framework is open-source and built entirely with free data and standard Python libraries.

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A Supplementary Figures



FI-PPO: Factor-Informed PPO Architecture

$$\mathcal{L}^{\text{FI-PPO}} = \mathcal{L}^{\text{PPO}} + \lambda_{IC} \mathcal{L}_{IC} + \lambda_{ortho} \mathcal{L}_{ortho}$$

Figure 6: FI-PPO system architecture.

Figure 6: Training Dynamics

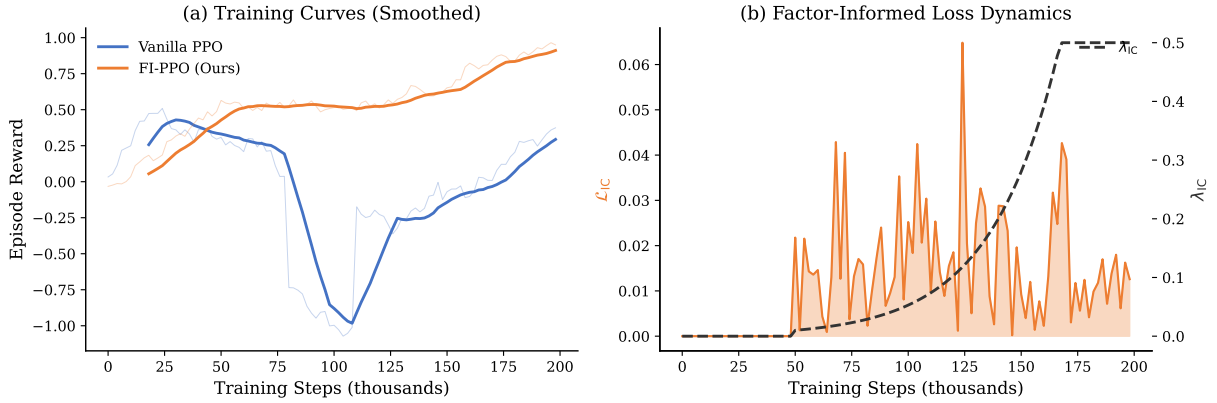


Figure 7: Training dynamics: (a) Episode reward curves for vanilla PPO and FI-PPO; (b) Factor-informed loss contribution and curriculum schedule.

Figure 7: Hierarchical State Space Decomposition

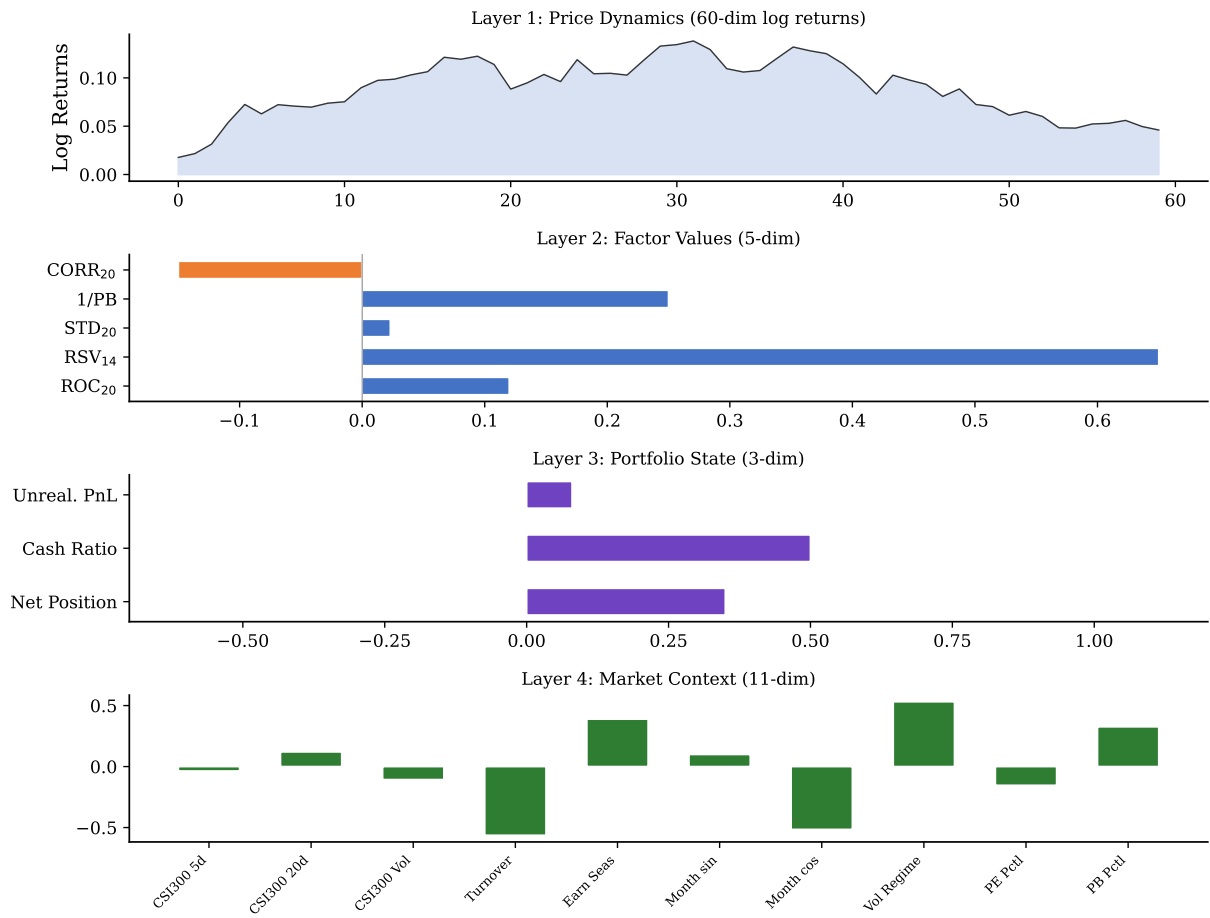


Figure 8: Hierarchical state space decomposition.

Figure 9: Equity Curves — FI-PPO vs Vanilla PPO vs Buy & Hold (2022–2025)

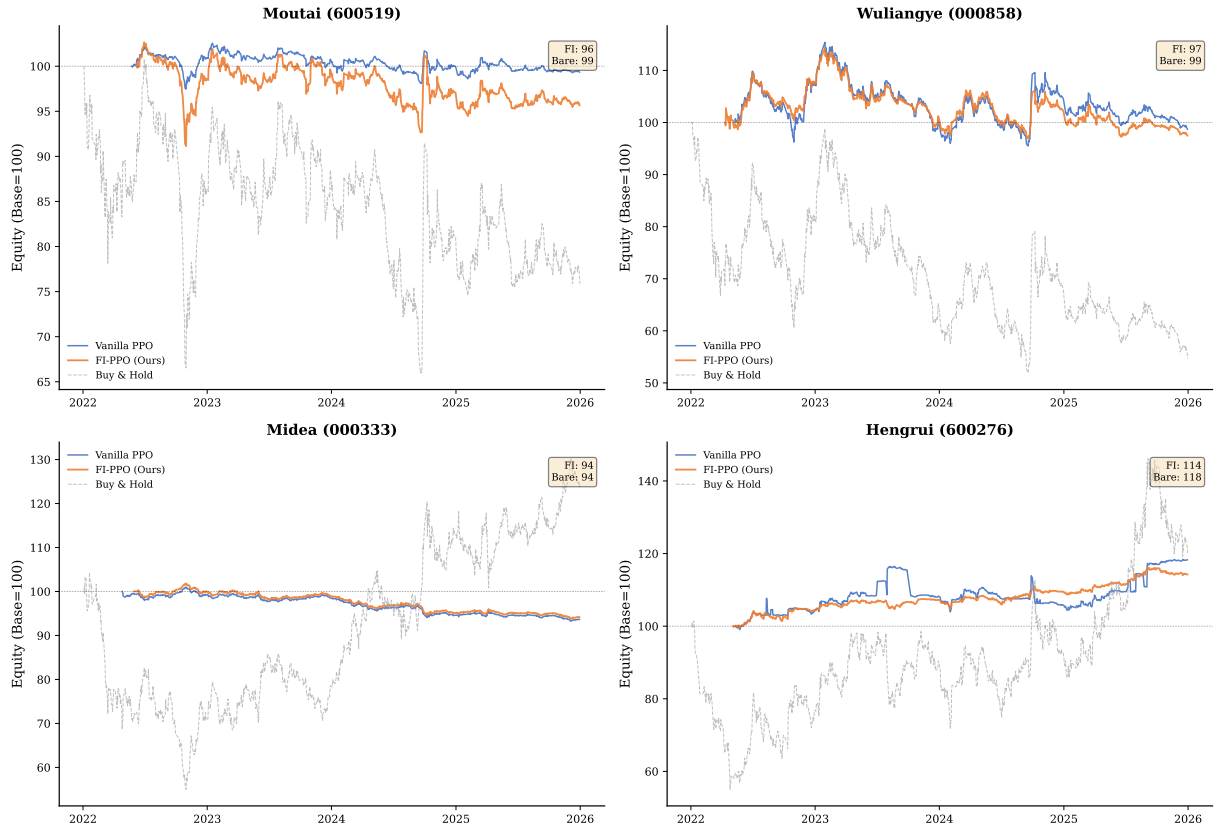


Figure 9: Equity curves for four core stocks (2022–2025 test period). Each panel compares FI-PPO (orange), vanilla PPO (blue), and Buy&Hold (gray dashed). Portfolio value is normalized to 100 at the start of the test period.

Figure 10: Position Dynamics — FI-PPO vs Vanilla PPO

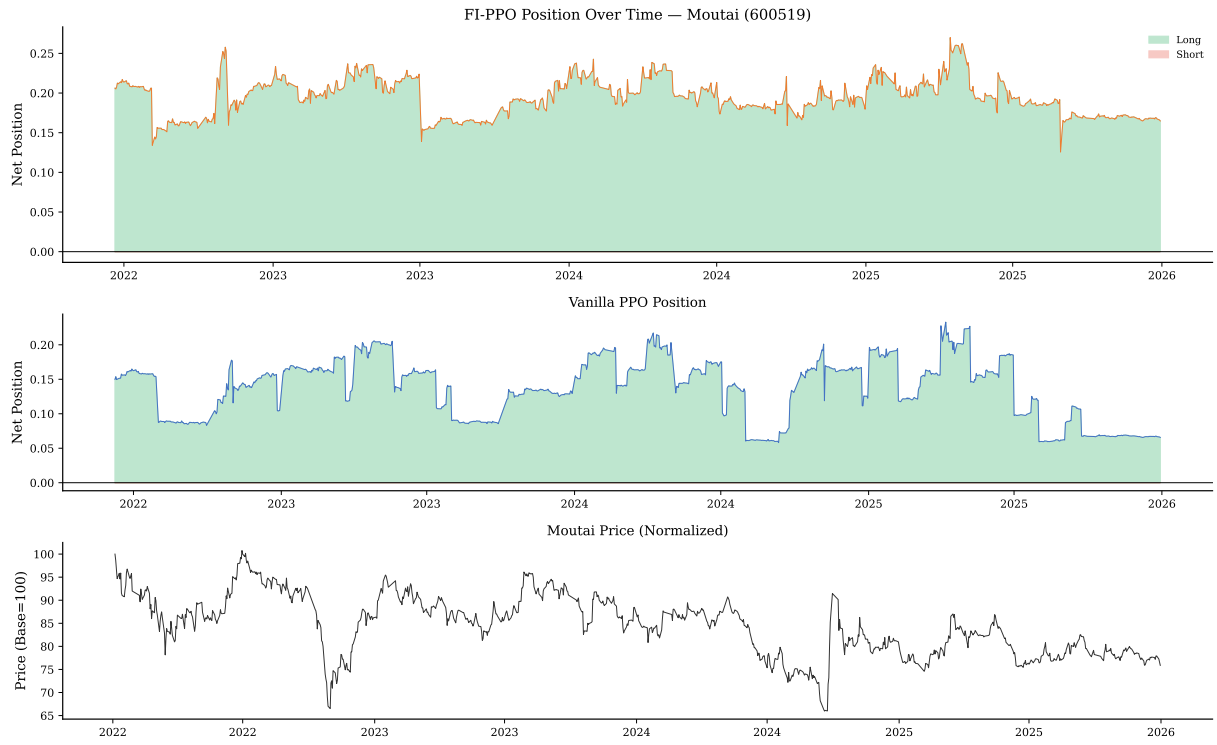


Figure 10: Position dynamics for Kweichow Moutai (600519). Top: FI-PPO net position showing active long/short management. Middle: Vanilla PPO net position. Bottom: Normalized stock price for context. Green shading indicates long exposure; red shading indicates short exposure.